Note 2.5 Cardinality of Sets

Introduction

In this section, we will know about the cardinality of infinite sets.

Definition of the Size of Cardinality

The sets A and B have the same cardinality if and only if there is a one-to-one correspondence from A to B, denoted by $|A| = |B|$.
If there is a one-to-one function from A to B, the cardinality of A is less than or the same as the cardinality of B and we write $|A| \leq |B|$. If they have different cardinality then it is $|A| < |B|$.

Countable Sets

Definition of Countable Sets

A set that is either finite or has the same cardinality as the set of positive integers is called countable. A set that is not countable is called uncountable. If a set is countable, we denote the cardinality of S by $\aleph_0$, reading as "aleph null".

The set of all integers and all rational numbers are countable.

An Uncountable Set

The set of all real numbers are uncountable.

Results about Cardinality

Theorem 1

If A and B are countable sets, then $A \cup B$ is also countable.

Theorem 2--SCHRODER-BERNSTEIN THEOREM

If A and B are sets with $|A| \leq |B|$ and $|B| \leq |A|$, then $|A| = |B|$. In other words, if there are one-to-one functions f from A to B and g from B to A, then there is a one-to-one correspondence between A and B.

Uncomputable Functions

We say that a function is computable if there is a computer program in some programming language that finds the values of this function. If not we say it is uncomputable.